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WORK EXPERIENCE

Dec 2023- Present	UNIVERSITY OF CINCINNATI , CINCINNATI, USA <i>Assistant Professor</i> , Department of Aerospace Engineering and Engineering Mechanics
Dec 2021- Nov 2023	THE UNIVERSITY OF TEXAS AT SAN ANTONIO , SAN ANTONIO, USA <i>Postdoctoral Fellow</i> , Host : Dr. Yongcan Cao
May 2021- Nov 2021	INDIAN INSTITUTE OF TECHNOLOGY BOMBAY , MUMBAI, INDIA <i>Postdoctoral Fellow</i> , Host : Dr. Shashi Ranjan Kumar
Jun 2019- May 2018	CENTRAL SCIENTIFIC INSTRUMENTS ORGANISATION , CHANDIGARH, INDIA <i>Research Student</i>
Sep 2014- Jul 2016	TATA CONSULTANCY SERVICES LIMITED , PUNE, INDIA <i>Assistant System Engineer</i> , Consulting in Manufacturing, Automation and Control Systems
Nov 2015	TATA CONSULTANCY SERVICES LIMITED , TRIVANDRUM, INDIA <i>Visiting faculty at TCS Global Learning Center</i>
Jun 2011- May 2014	KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY , BHUBANESWAR, INDIA <i>Instructor and Coordinator</i> , KIIT Robotics Society

EDUCATION

 INDIAN INSTITUTE OF TECHNOLOGY BOMBAY Doctor of Philosophy (Ph.D.), Aerospace Engineering	2018 - 2021
 INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY Master of Technology (M.Tech.), Mechatronics	2016 - 2018
 KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY Bachelor of Technology (B.Tech.), Electronics and Instrumentation	2010 - 2014

PATENTS

- [P1] Y. Cao, D. Casbeer, P. K. Ranjan, **Abhinav Sinha**, and I. Weintraub, "System and Method for Flexible Relational Maneuvering of Leader-Follower Unmanned Aerial Vehicles," U.S. Patent US20260029804A1 [Pending]

PUBLICATIONS

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- [J36] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, "Three-dimensional Integrated Guidance and Control for Leader-Follower Flexible Formation of Fixed Wing UAVs", *Aerospace Science and Technology*, 2026 [to appear]

- [J35] S. Bajpai, **Abhinav Sinha**, and S. R. Kumar, "Cooperative Guidance for Aerial Defense in Multiagent Systems", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 3700–3710, 2026.
- [J34] **Abhinav Sinha**, and S. R. Kumar, "Cooperative Target Interception in 3D Engagements Over Switched Dynamic Graphs", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 4181–4192, 2026.
- [J33] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, "Three-dimensional Nonlinear Path-following Guidance with Bounded Input Constraints", *AIAA Journal of Guidance, Control, and Dynamics*, vol. 49, no. 1, pp. 198–215, 2026.
- [J32] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, "Three dimensional Trajectory Tracking for Unmanned Aerial Vehicles under Motion Constraints", *AIAA Journal of Guidance, Control, and Dynamics*, vol. 49, no. 1, pp. 78–96, 2026.
- [J31] U. Siddique, **Abhinav Sinha**, and Y. Cao, "Learning Fair Policies in Multi-Objective Preference-based Reinforcement Learning", *Machine Learning Journal (MLJ)*, vol. 115, no. 23, pp. 1–34, 2026.
- [J30] **Abhinav Sinha**, D. Mukherjee, and S. R. Kumar, "Consensus-driven Deviated Pursuit for Guaranteed Simultaneous Interception of Moving Targets", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 5, pp. 12826–12838, 2025.
- [J29] P. Ranjan, **Abhinav Sinha**, and Y. Cao, "3D Guidance Law for Flexible Target Enclosing with Inherent Safety", *IEEE Robotics and Automation Letters*, vol. 10, no. 2, pp. 2088–2095, 2025.
- [J28] P. Ranjan, **Abhinav Sinha**, and Y. Cao, "Self-organizing Multiagent Target Enclosing under Limited Information and Safety Guarantees", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 1, pp. 1066–1078, 2025.
- [J27] P. Ranjan, **Abhinav Sinha**, Y. Cao, D. Casbeer, and I. Weintraub, "Relational Maneuvering of Leader-Follower Unmanned Aerial Vehicles for Flexible Formation", *IEEE Transactions on Cybernetics*, vol. 54, no. 10, pp. 5598–5609, 2024.
- [J26] **Abhinav Sinha** and S. R. Kumar, "Time-Critical Unified Rendezvous Guidance for an Unmanned Autonomous Vehicle", *ASME Journal of Autonomous Vehicles and Systems*, vol. 4, no. 3, pp. 031001 1–8, 2024.
- [J25] S. Kumar, **Abhinav Sinha**, and S. R. Kumar, "Robust Path-following Guidance for an Autonomous Vehicle in the Presence of Wind", *Aerospace Science and Technology*, vol. 150, pp. 109225, 2024.
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- [J20] **Abhinav Sinha** and Y. Cao, "3-D Nonlinear Guidance Law for Target Circumnavigation", *IEEE Control System Letters*, vol. 7, pp. 655–660, 2023 (also accepted for presentation at 2023 American Control Conference, San Diego, California, USA).
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- [J18] **Abhinav Sinha** and Y. Cao, "Nonlinear Guidance Law for Target Enclosing with Arbitrary Smooth Shapes", *AIAA Journal of Guidance, Control, and Dynamics*, vol. 45, no. 11, pp. 2182–2192, 2022.
- [J17] **Abhinav Sinha** and S. R. Kumar, "Cooperative target capture using predefined-time consensus over fixed and switching networks", *Aerospace Science and Technology*, vol. 127, pp. 107686, 2022.

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- [J15] **Abhinav Sinha**, S. R. Kumar, and D. Mukherjee, “Cooperative integrated guidance and control design for simultaneous interception”, *Aerospace Science and Technology*, vol. 120, pp. 107262, 2022.
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- [J10] **Abhinav Sinha**, S. R. Kumar, and D. Mukherjee, “Three-Dimensional Guidance With Terminal Time Constraints for Wide Launch Envelops”, *AIAA Journal of Guidance, Control, and Dynamics*, vol. 44, no. 2, pp. 343–359, 2021.
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- [J5] R. K. Mishra, and **Abhinav Sinha**, “Event-triggered sliding mode based consensus tracking in second order heterogeneous nonlinear multi-agent systems”, *European Journal of Control*, vol. 45, pp. 30–44, 2019.
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- [J2] T. Majumder, R. K. Mishra, **Abhinav Sinha**, S. S. Singh, and P. K. Sahu, “Congestion Control in Cognitive Radio Networks with Event-triggered Sliding Mode”, *AEU– International Journal of Electronics and Communication*, vol. 90, pp. 155–162, 2018.
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- [j5] **Abhinav Sinha** and S. R. Kumar, “Distributed Safe Consensus Under Asymmetric Input and Time-Varying Output Constraints”
- [j4] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, “Provably Safe Control for Constrained Nonlinear Systems with Bounded Input”

- [j3] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Predefined-time One-Shot Cooperative Estimation, Guidance, and Control for Simultaneous Target Interception”
- [j2] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, “Engagement-Zone-Aware Input-Constrained Guidance for Safe Target Interception in Contested Environments”
- [j1] S. Bajpai and **Abhinav Sinha**, “Terminal Time and Angle-Constrained Nonlinear Intercept Guidance”

BOOK CHAPTERS

- [BC2] **Abhinav Sinha**, and R. K. Mishra, “Smooth sliding mode control of a nonlinear CSTR using an inverse hyperbolic function-based law, *Foundations and Frontiers in Computer, Communication and Electrical Engineering*, Taylor and Francis, 2016.
- [BC1] T. Majumder, **Abhinav Sinha**, R. K. Mishra, S. S. Singh, and P. K. Sahu, “Congestion control in Cognitive Radio Networks using fractional order rate reaching law based sliding modes, *Foundations and Frontiers in Computer, Communication and Electrical Engineering*, Taylor and Francis, 2016.

PEER REVIEWED CONFERENCE PROCEEDINGS

- [C63] S. Bajpai and **Abhinav Sinha**, “Nonlinear Guidance Law for Terminal Time and Angle-Constrained Interception”, in *Proc., Modeling, Estimation, and Control Conference (MECC)*, Arizona, USA, 25th-28th Oct, 2026.
- [C62] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, “GPS-Denied Guidance Strategy for Safe Target Interception Under Multi-Defender Threats”, in *Proc., Modeling, Estimation, and Control Conference (MECC)*, Arizona, USA, 25th-28th Oct, 2026.
- [C61] U. Siddique, P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, “Deep Reinforcement Learning for Target Enclosing Under Limited Sensing and Autopilot Lag”, in *Proc., Modeling, Estimation, and Control Conference (MECC)*, Arizona, USA, 25th-28th Oct, 2026.
- [C60] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Design of Exact-Time Convergent Impact Angle Guidance Laws using Trigonometric Sliding Modes”, in *Proc., 18th International Workshop on Variable Structure Systems and Sliding Mode Control (VSS)*, Exeter, UK, 27th-30th Jul, 2026.
- [C59] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Nonlinear Cooperative Salvo Guidance with Seeker-Limited Interceptors”, in *Proc., 18th International Workshop on Variable Structure Systems and Sliding Mode Control (VSS)*, Exeter, UK, 27th-30th Jul, 2026.
- [C58] R. V. Nanavati, **Abhinav Sinha**, and S. R. Kumar, “Nonlinear Control of a Fully-Actuated UAV”, in *Proc., 18th International Workshop on Variable Structure Systems and Sliding Mode Control (VSS)*, Exeter, UK, 27th-30th Jul, 2026.
- [C57] **Abhinav Sinha**, and R. V. Nanavati, “Robust Near-Optimal Nonlinear Target Enclosing Guidance”, in *Proc., 18th International Workshop on Variable Structure Systems and Sliding Mode Control (VSS)*, Exeter, UK, 27th-30th Jul, 2026.
- [C56] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Cooperative Integrated Estimation-Guidance for Simultaneous Interception of Moving Targets”, in *Proc., European Control Conference (ECC)*, Reykjavík, Iceland, 7th-10th Jul, 2026.
- [C55] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Bounded-Input True Proportional Navigation for Impact-Time Control”, in *Proc., Asian Control Conference (ASCC)*, Bali, Indonesia, 17th-21st Jun, 2026.
- [C54] **Abhinav Sinha**, D. Mukherjee, and S. R. Kumar, “On Robustness of Consensus over Pseudo-Undirected Path Graphs”, in *Proc., American Control Conference (ACC)*, New Orleans, USA, 26th-29th May, 2026.
- [C53] L. Gopikannan, S. R. Kumar, and **Abhinav Sinha**, “Trajectory Encryption Cooperative Salvo Guidance”, in *Proc., American Control Conference (ACC)*, New Orleans, USA, 26th-29th May, 2026.
- [C52] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, “Safety-Critical Input-Constrained Nonlinear Intercept Guidance in Multiple Engagement Zones”, in *Proc., American Control Conference (ACC)*, New Orleans, USA, 26th-29th May, 2026.

- [C51] U. Siddique, **Abhinav Sinha**, and Y. Cao, "Adaptive Event-Triggered Policy Gradient for Multi-Agent Reinforcement Learning", in *Proc., American Control Conference (ACC)*, New Orleans, USA, 26th-29th May, 2026.
- [C50] S. Bajpai, **Abhinav Sinha**, I. Weintraub, R. Sharma, and D. Casbeer, "Nonlinear Guidance for a Pursuer-Turret Team for Maneuvering Target Engagement", in *Proc., American Control Conference (ACC)*, New Orleans, USA, 26th-29th May, 2026.
- [C49] S. Bajpai, P. Sood, **Abhinav Sinha**, and R. Sharma, "Terminal Time and Angle-constrained Interception of a Non-Maneuvering Target under Autopilot Lag", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C48] S. Bajpai, **Abhinav Sinha**, and R. Sharma, "Cooperative Defense via Region-Based Interception in Turret-Evader-Pursuer Engagement", in *Proc., Intelligent Systems Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C47] R. Boyinine and **Abhinav Sinha**, "Observability-aided Target Interception using Bearing-only Measurements via Cooperative Estimation", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C46] R. Sharma, **Abhinav Sinha**, and I. Weintraub, "Learning-Augmented Adaptive Guidance for Target Interception", in *Proc., Intelligent Systems Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C45] **Abhinav Sinha**, R. V. Nanavati, and S. R. Kumar, "Impact Time Guidance to Capture Moving Targets in the Presence of Multiple Obstacles", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C44] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, "Nonlinear Path Following Guidance with Bounded Control", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C43] R. Tabiyar, **Abhinav Sinha**, and S. R. Kumar, "3D Cooperative Salvo against a Stationary Target with Arbitrary Time Consensus", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 12th-16th Jan, 2026.
- [C42] U. Siddique, **Abhinav Sinha**, and Y. Cao, "Adaptive Event-triggered Reinforcement Learning Control for Complex Nonlinear Systems", in *Proc., American Control Conference (ACC)*, Denver, USA, 8th-10th Jul, 2025.
- [C41] R. Boyinine, A. Chakraborty, R. Sharma, and **Abhinav Sinha**, "Infrastructure Planning for Multi-Vehicle Routing with Cooperative Localization", in *Proc. 2025 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, Salt Lake City, USA, 28th Apr-1st May, 2025.
- [C40] S. Bajpai, R. Boyinine, **Abhinav Sinha**, and R. Sharma, "Toward Simultaneous Localization and Guidance for Collaborative Combat Aircraft (CCA) Effectiveness", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 6th-10th Jan, 2025.
- [C39] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, "Geometric Guidance for Enclosing Moving Targets", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 6th-10th Jan, 2025.
- [C38] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, "Relational Maneuvering Guidance Strategy for Leader-Follower Flexible Formation in Three Dimensions", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 6th-10th Jan, 2025.
- [C37] S. Bandela, **Abhinav Sinha**, and Y. Cao, "A Reinforcement Learning Approach to Target Capture in the Presence of a Defender", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 6th-10th Jan, 2025.
- [C36] M. Wu, U. Siddique, **Abhinav Sinha**, and Y. Cao, "Offline Reinforcement Learning with Failure Under Sparse Reward Environments", in *Proc., 3rd International Conference on Computing and Machine Intelligence (ICMI)*, Mt. Pleasant (Michigan), USA, 13th-14th Apr, 2024.
- [C35] U. Siddique, **Abhinav Sinha**, and Y. Cao, "On Deep Reinforcement Learning for Target Capture Autonomous Guidance", in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 8th-12th Jan, 2024.

- [C34] P. K. Ranjan, **Abhinav Sinha**, and Y. Cao, “Robust UAV Guidance Law for Safe Target Circumnavigation with Limited Information and Autopilot Lag Considerations”, in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 8th-12th Jan, 2024.
- [C33] **Abhinav Sinha**, D. White, and Y. Cao, “Deep Reinforcement Learning-based Optimal Time-constrained Intercept Guidance”, in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 8th-12th Jan, 2024.
- [C32] S. Kumar, **Abhinav Sinha**, and S. R. Kumar, “Generic Path-following Guidance for an Autonomous Vehicle”, in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 8th-12th Jan, 2024.
- [C31] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, “Robust Nonlinear Control for Exact-time Stability of a Quadrotor UAV under Uncertainties”, in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, Orlando, USA, 8th-12th Jan, 2024.
- [C30] R. Tabiyar, **Abhinav Sinha**, and S. R. Kumar, “Decentralized Cooperative Guidance for Time-constrained Rendezvous with Non-Accelerating Targets”, in *Proc., 9th Indian Control Conference (ICC)*, Vishakhapatnam, India, 18th-20th Dec, 2023.
- [C29] U. Siddique, **Abhinav Sinha**, and Y. Cao, “Fairness in Preference-based Reinforcement Learning”, in *41st International Conference on Machine Learning (ICML) Workshop : The Many Facets of Preference-Based Learning*, Honolulu, USA, 26th-27th Jul, 2023.
- [C28] **Abhinav Sinha** and Y. Cao, “Three-dimensional Nonlinear Guidance Law for Target Circumnavigation”, in *Proc., American Control Conference (ACC)*, San Diego, USA, 31st May-2nd Jun, 2023.
- [C27] **Abhinav Sinha** and Y. Cao, “A Nonlinear Guidance Law for Target Enclosing with Arbitrary Smooth Shapes”, in *Proc., Guidance, Navigation, and Control Conference (AIAA SciTech Forum)*, National Harbor, USA, 23rd-27th Jan, 2023.
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- [C10] **Abhinav Sinha**, and R. K. Mishra, "Convergence of multi-agent systems to unknown source of an odor", in *Proc., 2018 IEEE 3rd International Conference for Convergence in Technology*, Pune, India, 7th-8th Apr 2018.
- [C9] **Abhinav Sinha**, and R. K. Mishra, "Temperature regulation in a Continuous Stirred Tank Reactor using event triggered sliding mode control", *IFAC PapersOnLine*, vol. 51, no. 1, pp. 401–406, 2018.
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- [C6] **Abhinav Sinha**, and R. K. Mishra, "Robust altitude tracking of a miniature helicopter UAV based on sliding mode", in *Proc., 2015 IEEE International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS)*, Coimbatore, India, 19th-20th Mar 2015.
- [C5] T. Majumder, **Abhinav Sinha**, R. K. Mishra, S. S. Singh, and P. K. Sahu, "Robust nonlinear congestion controller for cognitive radio based wireless network", in *Proc., 2015 IEEE International Conference on Innovations in Information, Embedded and Communication Systems (ICIIECS)*, Coimbatore, India, 19th-20th Mar 2015.
- [C4] **Abhinav Sinha**, R. K. Mishra, and S. Jaiswal, "Robust and Smooth Nonlinear Control of an Industrial Robot for Automated Pick and Place", in *Proc., 2015 IEEE International Conference on Computing Communication Control and Automation (ICCUBEA)*, Pune, India, 26th-27th Feb 2015, **Awarded Best Paper**.
- [C3] **Abhinav Sinha**, P. Prasoon, P. K. Bharadwaj, and A. C. Ranasinghe, "Nonlinear Autonomous Control of a Two-Wheeled Inverted Pendulum Mobile Robot Based on Sliding Mode", in *Proc., 2015 IEEE International Conference on Computational Intelligence and Networks (CINE)*, Bhubaneswar, India, 12th-13th Jan 2015.
- [C2] A. C. Ranasinghe, K. Rasnayake, **Abhinav Sinha**, and K. K. Rasnayake, "Perturbing effect compensation technique for smart sensors", in *Proc., 7th IEEE International Conference on Information and Automation for Sustainability (ICIAfS)*, Colombo, Sri Lanka, 22nd-24th Dec 2014.
- [C1] **Abhinav Sinha**, and R. K. Mishra, "Smooth sliding mode controller design for robotic arm", in *Proc., 2013 International Conference on Control, Automation, Robotics and Embedded Systems (CARE)*, Jabalpur, India, 16th-18th Dec 2013.

OTHERS

- [A4] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, “Generalization of Path Following Problem : A Guidance Perspective”, in *Autonomous Systems and Emerging Technologies Symposium (ASSETS)*, Bangalore, India, 25th-27th Jun 2025.
- [A3] S. Kumar, S. R. Kumar, and **Abhinav Sinha**, “Three-Dimensional Nonlinear Path-Following Guidance for UAVs”, in *8th Cyber-Physical Systems Symposium (CyPhySS)*, Bangalore, India, 25th-27th Jul 2024.
- [A2] **Abhinav Sinha**, S. R. Kumar, and D. Mukherjee, “Integrated Guidance and Control Design For Time-Constrained Interception”, *IEEE TechRxiv*, 2020.
- [A1] **Abhinav Sinha**, R. Kaur, R. Kumar, and A. P. Bhondekar, “Cooperative control of multi-agent systems to locate source of an odor”, *arXiv:1711.03819, arXiv e-prints*, 2017.

GRANTS

- [G2] 2025-2027 : Principal Investigator, **AIR-GAME CCA : Autonomous Information-Driven Response for Guided Autonomous Missions and Engagement in Collaborative Combat Aircraft**, The US Air Force Research Labs
- [G1] 2024-2025 : Principal Investigator, **CO AIR : Cooperative Operation of Assets using Intelligent Reinforcements to Enhance the Collaborative Combat Aircraft Capability**, UtopiaCompression Corporation and AFWERX

COURSES TAUGHT/OFFERED

- AEEM 5115/6015 : Modern Control (S 25, F 26–)
- AEEM 3015 : Fundamentals of Control Theory (S 26–)
- AEEM 4042 : Fundamentals of Control Theory (F 24, 25)
- AEEM 5015C/5025C : UAV Design (every F and S)
- AEEM 9074 : Introduction to Navigation and Guidance
- AEEM 9074 : Control of Networked Cyber-Physical Systems
- AEEM 9074 : Reinforcement Learning and Data-driven Control
- AEEM 2013 : Introduction to Systems Engineering (S 24)

INVITED TALKS, SEMINARS, TUTORIALS

- [T4] Mar 2026 : **Provably Safe Nonlinear Control Design for a Class of Nonlinear Systems**, Short Term Course cum Faculty Development Program on Trends and Challenges in Cyber-Physical Energy Systems (TCCPES-2026), NIT Rourkela
- [T3] Dec 2025 : **Provably Safe Nonlinear Control Design for a Class of Nonlinear Systems**, UC-AEEM Graduate Seminar Series
- [T2] Jan 2025 : **On the Lady, the Bandit, and the Bodyguard Problem : Cooperative Guidance Strategies for Guaranteed Pursuit-Evasion**, UC-AEEM Graduate Seminar Series
- [T1] Jan 2025 : **On the Lady, the Bandit, and the Bodyguard Problem : Cooperative Guidance Strategies for Guaranteed Pursuit-Evasion**, Short Term Course cum Faculty Development Program on Advances in Control, Estimation & Optimization For Cyber-Physical Systems (ACEOCPS), NIT Rourkela

STUDENTS ADVISED/MENTORED, STAFFS

POSTDOCTORAL RESEARCH FELLOW

- [P1] Dr. Rohith Boyinine (2025–2026), next : Samsung Semiconductor, California

DOCTORAL STUDENTS

- [D2] Abhimanyu Khadga (2025–)
- [D1] Shivam Bajpai (2024–)

MASTERS STUDENTS

- [M5] Sandeep Bandrupalli (2024–), currently intern at LeanQubit Inc., Ohio
- [M4] Emilio Herrera (2026), UC RIAS programme
- [M3] Mukhtar Ahmed (2025)
- [M2] Liam McKenna (2025), next : Air Force Institute of Technology, WPAFB, Ohio
- [M1] Jayanth Ammapalli (2025), next : UtopiaCompression Corporation, California

BACHELOR STUDENTS

- [B2] Pragyat Sood (2024– 2025)
- [B1] Hari Narayanan Sathyanarayanan (2024)

PROFESSIONAL SERVICES

EDITORIAL POSITIONS

- Associate Editor (Guidance and Control Systems), IEEE Transactions on Aerospace and Electronic Systems
- Associate Editor, IEEE CSS Conferences Editorial Board (ACC, CDC, CCTA)
- Guest Associate Editor (Special Session on Safety, Robustness, and Effectiveness in Human-Machine Teaming), IEEE Control Systems Letters
- Track Chair (Control, Robotic, Network, and System Sciences and Technologies), 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

REVIEWER OF REFEREED JOURNALS

- IEEE Transactions on Automatic Control
- Automatica
- International Journal of Control
- IEEE Transactions on Control Systems Technology
- International Journal of Robust and Nonlinear Control
- Systems and Control Letters
- IEEE Transactions on Control of Network Systems
- AIAA Journal of Guidance, Control, and Dynamics
- IEEE Control Systems Letters
- IEEE Robotics and Automation Letters
- IEEE Transactions on Aerospace and Electronic Systems
- Journal of Intelligent & Robotic Systems
- Aerospace Science and Technology
- IEEE Transactions on System, Man and Cybernetics : Systems
- IEEE Transactions on Cybernetics
- Journal of The Franklin Institute
- European Journal of Control
- IET Control Theory and Applications
- IEEE Transactions on Fuzzy Systems
- IEEE Transactions on Neural Networks and Learning Systems
- Asian Journal of Control
- Information Sciences
- ASME Journal of Dynamic Systems, Measurement, and Control
- IEEE Transactions on Industrial Electronics
- IEEE Transactions on Industrial Informatics
- ISA Transactions
- Journal of Process Control
- IEEE Access
- Defence Technology

- IEEE Systems Journal
- IEEE Sensors Letters
- Nonlinear Dynamics
- Advances in Space Research
- IET Generation, Transmission and Distribution
- Chemical Engineering Science
- Asia Pacific Journal of Chemical Engineering
- Journal of Computational Science
- Neurocomputing
- Swarm Intelligence
- IET Power Electronics
- Journal of Aerospace Engineering
- IEEE Journal of Emerging and Selected Topics in Industrial Electronics
- Plos One
- The Aeronautical Journal
- The Journal of the Astronautical Sciences
- IEEE Journal of Oceanic Engineering
- IEEE Transactions on Network Science and Engineering
- IEEE Open Journal of Control Systems
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Intelligent Vehicles

REVIEWER OF PEER REVIEWED CONFERENCES (SELECTED)

- American Control Conference (ACC)
- Control and Decision Conference (CDC)
- Indian Control Conference (ICC)
- European Control Conference (ECC)
- AIAA SciTech Forum
- IFAC World Congress
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Workshop on Variable Structure Systems and Sliding Mode Control (VSS)
- International Conference on Control, Decision and Information Technologies (CoDIT)
- International Conference on Control and Automation (ICCA)
- Mediterranean Conference on Control and Automation (MED)
- International Conference on Unmanned Aircraft Systems (ICUAS)
- IFAC Advances in Control and Optimization of Dynamical Systems (ACODS)
- IFAC Symposium on Automatic Control in Aerospace (ACA)
- IEEE Conference on Control Technologies and Applications (CCTA)
- International Conference on Automation and Computing

ACTIVITIES OF TECHNICAL COMMITTEES

- Vice Chair (2026-Present) of the IEEE CSS TC on Manufacturing Automation and Robotic Control
- Sub-committee Chair (Membership/Website, 2023-2026) of the IEEE CSS TC on Manufacturing Automation and Robotic Control
- Co-organizer (with S. Maity and R. K. Mishra, NIT Rourkela, and H. Wan, Jiangnan University) of Session on **Distributed Coordinated Control for Multi-Agent Systems** at the Asian Control Conference (ASCC), Bali, Indonesia, 17th-21st Jun 2026.
- Co-chair (and co-organizer with Isaac Weintraub, and Alexander Von-Moll) of Invited Session on **Multi-Agent Control and Coordination II** at the American Control Conference (ACC), New Orleans, USA, 26th-29th May 2026.

- Chair (and co organizer with Isaac Weintraub, Rajnikant Sharma, and Alexander Von-Moll) of Session on **IS-10 : Multi-Agent Control and Coordination I** at the AIAA 2026 SciTech Forum, Orlando, USA, 12th Jul-16th Jan 2026.
- Chair of Session on **GNC-26/IS-24 : Guidance, Navigation, and Control in Intelligent Systems** at the AIAA 2025 SciTech Forum, Orlando, USA, 6th Jul-10th Jan 2025.
- Chair and Organizer (co-organizer : Yongcan Cao, UTSA) of Invited Session on **Synergistic Strategies for Cyber-Physical-Human Systems** at the American Control Conference (ACC), Denver, USA, 8th Jul-10th Jul 2025.
- Organizer (co-organizers : Yongcan Cao, UTSA and Xiaocong Li, Harvard University/A*STAR Singapore) of Session on **Control Technologies and Solutions for Sustainable Energy Systems** at the IEEE CSS Day 2024.
- Chair and Organizer (co-organizers : Yongcan Cao, UTSA and David Casbeer, AFRL) of Invited Session on **Control of Cyber-Physical Systems : Multidisciplinary Approaches in Robotics, Autonomy, Optimization, and Safety** at the American Control Conference (ACC), Toronto, Canada, 8th Jul-12th Jul 2024.
- Organizer (co-organizers : Yongcan Cao, UTSA and Eloy Garcia, AFRL) of Invited Session on **Methods in Robotics, Optimization, Learning, and Safety for Control of Cyber-Physical Systems** at the American Control Conference (ACC), San Diego, USA, 31st May-2nd Jun 2023.
- Co-chair, Session on **Aerospace** at the American Control Conference (ACC), San Diego, USA, 31st May-2nd Jun 2023.



PROFESSIONAL MEMBERSHIPS

- **Life Member**, IITB Alumni Association
- **Senior Member**, Institute of Electrical and Electronics Engineers (IEEE)
 - ➔ IEEE Aerospace and Electronic Systems Society (AESS)
 - ➔ IEEE Control Systems Society (CSS)
 - ➔ IEEE Robotics and Automation Society (RAS)
 - ➔ **Vice Chair**, IEEE CSS Technical Committee on Manufacturing, Automation, and Robotic Control
 - ➔ IEEE CSS Technical Committee on Aerospace Controls
 - ➔ IEEE CSS Technical Committee on Automotive Controls
 - ➔ IEEE CSS Technical Committee on Control Education
- **Senior Member**, American Institute of Aeronautics and Astronautics (AIAA)
- **Member**, International Federation of Automatic Control (IFAC)
 - ➔ IFAC Automatic Control and Dynamic Optimization Society (ACDOS)
- **Member**, Asian Control Association



AWARDS, ACHIEVEMENTS AND HONORABLE MENTIONS

- **EXCELLENT JGCD REVIEWER 2024-25**
American Institute of Aeronautics and Astronautics, USA

DEC 2025
- **NAIK AND RASTOGI AWARD FOR EXCELLENCE IN PH.D. RESEARCH**
Indian Institute of Technology Bombay, India

AUG 2022
- **ACC STUDENT TRAVEL GRANT**
American Automatic Control Council, USA

JUN 2020, MAY 2021
- **ICC STUDENT SUPPORT**
Indian Control Conference and IEEE-CSS

DEC 2019
- **MHRD POSTGRADUATE GATE FELLOWSHIP FOR M.TECH, PH.D.**
Ministry of Human Resource Development, India

JUL 2016 - APR 2021
- **CHAMPIONS OF INITIAL LEARNING PROGRAMME (ILP)**
TATA Consultancy Services Limited

DEC 2015
- **ON THE SPOT AWARD**
TATA Consultancy Services Limited

JUL 2015
- **OUTSTANDING CONTRIBUTION AWARD**
KIIT Robotics Society, Kalinga Institute of Industrial Technology

AUG 2013